Assignment 4

Ling 201 – Fall 2026

(100 points total)

Part 1. Truth Values

For each statement below, **indicate** whether it would be true or false in World A and World B, as defined above.

Statement	**World A**	**World B**
1. Quartz is running.	T	F
2. Vincent is not sleeping.	F	T
3. Junwoo is a friendly reader.	T	F
4. Fahad is a tired singer.	F	F
5. Sukchan is brave and singing.	T	F
6. Chenli is happy and running.	F	F
7. Merlin is friendly but not tired.	F	F
8. Beryl is singing and friendly.	T	T
9. Utku is a tired reader.	F	F
10. Tetsu is sleeping and happy.	F	F

Part 2. Entailment Relations I

For each pair of sentences below, determine **whether or not sentence (a) entails sentence (b)**, and **explain briefly why** in your own words.

1. (a) Mount Everest is taller than every other mountain.

   (b) Mount Everest is taller than Mount Kilimanjaro.

(a) entails (b)

If "Mount Everest is taller than every other mountain", then it must be taller than Mount Kilimanjoro as well. (b) is true whenever (a) is true, so (a) entails (b).

2. (a) Arthur went for a run at noon.

   (b) Arthur went for a run in the park at noon.

(a) does not entail (b)

If "Arthur went for a run at noon", it does not necessarily mean he runs in the park as well, (b) is not true whenever (a) is true, so (a) does not entail (b).

3. (a) Morgana said she studied for the test.

   (b) Morgana studied for the test.

(a) does not entail (b)

If "Morgana said she studied for the test", it does not necessarily have to be the truth, she might be lying, (b) is not true whenever (a) is true, so (a) does not entail (b).

4. (a) Lancelot ate nachos and enchiladas for dinner.

   (b) Lancelot ate enchiladas for dinner.

(a) entails (b)

If "Lancelot ate nochos and enchiladas for dinner", it must be the case that he ate enchiladas for dinner, (b) is true whenever (a) is true, so (a) entails (b).

5. (a) Mordred played a song on his guitar.

   (b) Mordred played music.

(a) entails (b)

If "Mordred played a song on his guitar", it must be the case that he played music, (b) is true whenever (a) is true, so (a) entails (b).

6. (a) Guinevere read a book yesterday.

   (b) Guinevere read a novel yesterday.

(a) does not entail (b)

If "Guinevere read a book yesterday", it is not necessarily that she read a novel, she might read a fiction or a poetry anthology, (b) is not true whenever (a) is true, so (a) does not entail (b).

Part 3. Entailment Relations II

For each pair of sentences below, identify the **entailment relations between two propositions** (*p* and *q*) and **justify your answer using semantic reasoning** (truth conditions, etc.).

a) *p*: Uther solved all questions on the exam.
   q: Uther solved some of the questions on the exam.

p entails *q*, but *q* does not entail *p*

If "Uther solved all questions on the exam", it must be the case that he solved part (some) of the questions on the exam, *q* is true whenever *p* is true, so *p* entails *q*.

If "Uther solved some of the questions on the exam", it is not necessarily that he solved all of the questions on the exam, *p* is not true whenever *q* is true, so *q* does not entail *p*.

b) *p*: Merlin broke the window.
q: The window was broken by Merlin.

p entails *q*, and *q* entails *p* (mutual entailment)

If "Merlin broke the window", it must be the case that the window was broken by Merlin, so *q* is true whenever *p* is true, so *p* entails *q*.

If "the window was broken by Merlin", it must be the case that Merlin broke the window, so *p* is true whenever *q* is true, so *q* entails *p*.

c) *p*: Gawain took the driving test.
q: Gawain failed the driving test.

p does not entail *q*, but *q* entails *p*

If "Gawain took the driving test", it is not necessarily that he failed the driving test, he could also pass the test, so *q* is not true whenever *p* is true, so *p* does not entail *q*.

If "Gawain failed the driving test", it must be the case that he took the driving test, *p* is true whenever *q* is true, so *q* entails *p*.

d) *p*: Perceval is reading a novel
q: Perceval is writing an essay.

p does not entail *q*, and *q* does not entail *p either*

If "Perceval is reading a novel", it is not necessarily that he is writing an essay, he could also do something else or do nothing, so *q* is not true whenever *p* is true, so *p* does not entail *q*.

If "Perceval is writing an essay", it is not necessarily that he is reading a novel, he could also do something else or do nothing, so *q* is not true whenever *p* is true, so *p* does not entail *q*.

Part 4. Identifying Entailment

For each of the following sentences, **write down two** more sentences that are entailed by that sentence. *(3 points/item, 18 points total)*

anything that is reasonable…